Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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Welcome to **Creating Electronic Books with Code and Artificial Intelligence Assistants**.

What is this?

This is the course material and a template designed for the teaching staff of the **Faculties of Sciences and Chemical Sciences at USAL** to create interactive teaching books easily. The main idea of this template is for it to be broad enough to cover many use cases and for each student to adapt it to their own course, while also being sufficiently equipped to be used easily with the help of AI assistants (such as GitHub Copilot, Gemini, Claude, Codex, etc.) and a code editor such as VS Code.

Content

In this book you will find:

- Tutorials to learn how to use the template
- Examples by Degree to see real cases
- Information on how to cite and licenses

PDF version

You can also download the printable version of the book:

- Download PDF in English
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> **Note**
>
> This project is designed to be used with **VS Code** and **AI** assistants.

Part I

Mathematics

QUADRATIC EQUATIONS

Quadratic equations have the general form:

$$ax^2 + bx + c = 0 \tag{1.1}$$

Where a, b, and c are constants and $a \neq 0$.

1.1 General formula

The solution to equation (1.1) is given by the general formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

CHAPTER 2

ORDINARY LEAST SQUARES (OLS)

The **Ordinary Least Squares** (OLS) method is a mathematical optimization technique used to find the line (or hyperplane) that best fits a set of data.

Its main objective is to minimize the sum of the squared differences (residuals) between the observed values and the values predicted by the model. That is, it searches for the linear regression coefficients that result in the smallest possible error.

2.1 Interactive Exploration

Below, you can manually adjust the parameters of a simple linear regression (the intercept β_0 and the slope β_1) to try to find the model that best fits the data, or allow the OLS algorithm to calculate the optimal value analytically.

Interactive visualization: Please refer to the HTML version of the book to access the Ordinary Least Squares (OLS) interactive tool.

Part II

Computer Science

CHAPTER 3

WEEK 3 EXERCISES: PROGRAMMING III

CONTENTS: Wrapper Classes and first OOP exercises

3.1 Mandatory Exercises:

1. You must create a `Person.java` class with attributes `name`, `height`, and `weight`, and their corresponding getters and setters. In the main method, you must instantiate 3 objects of this class and complete their information by asking the user via console for each object. Finally, you must show who is the tallest and who weighs the most. You must always use the getters and setters to get and set the information in each object instance.
2. Improve exercise 1 so that there is a parameterless constructor that initializes the values of name, height, and weight to default values that you consider appropriate.
3. Improve exercise 1 so that the `Person` class offers a method that calculates the person's BMI using the current values of the object's attributes. You must consider the possibility that the values have not been initialized to valid values.
4. Analyze the execution flow of the program by placing breakpoints in the constructor and in the different parts of the code of exercise 1. Set values to the attributes directly in their declaration and set a breakpoint to check what is executed first: the constructor or the attribute declaration.

3.2 Extension Exercises:

5. Using the `parseXXX` methods of the wrapper classes, build a class with appropriate methods to read all numeric data types (`int`, `float`, `double`) as strings. If the user does not enter a valid value, ask for the value again.
6. Modify exercise 1 so that the `Person.java` class is inside a package called `model` instead of the same package as the `App.java` class. What did you need to change to be able to use it in `App.java`?
7. What is the difference between the default constructor of a class in Java and a parameterless constructor?
8. Why use getters and setters and not access the attributes directly through the dot? Encapsulation. Is it a security or design issue?
9. What is the difference between defining an attribute as `public`, `private`, or no visibility modifier? In which case is it possible to access them through the dot (.) with an object reference?
10. What is the `this` keyword used for? And the `new` keyword?
11. It is possible to define classes that in turn have attributes of other classes (called Composition). You must now modify exercise 1 so that the `Person` class has an attribute of type `DNI` (another class that you will have to create) that will have as attributes number and letter. The constructor of the `Person` class must be passed a DNI

number as a parameter and an object of type `DNI` will be created, generating the corresponding letter from the algorithm to obtain the DNI letter from the number.

12. Create a class called `Rectangle.java` with length and width attributes, each with a default value of 1. It must have methods to calculate the perimeter and area of the rectangle. It must have getter and setter methods for length and width. The setter methods must verify that length and width are floating-point numbers greater than 0.0, and less than 20.0. Write a program to test the `Rectangle` class.

13. Create a new project with the `Person.java` class that you initially created. Perform the following instructions and answer the questions.

```java
public static void main(String[] args) {
    // Creation of an object of the Person class
    Person paco = new Person("Paco", 22, 177.3f, 82.3f);

    // We copy the reference of the previous object to another reference
    Person refPaco2 = paco;

    // We use the second reference to set a value in the object it references
    refPaco2.setAgeYears(77);

    // Result?
    System.out.printf("The age of %s is:%d%n", paco.getName(), paco.getAgeYears());

    // What is the age shown?
    // How many objects are in memory? How many references? Check the Debugger
}
```

Part III

Research

MULTIAGENT SYSTEM FOR CERVICAL EVALUATION

This chapter is based on the article *"A novel multiagent system for cervical motor control evaluation and individualized therapy: integrating gamification and portable solutions"* [MBR+24].

4.1 Study Overview

The study focused on designing a portable, objective device for assessing and addressing Cervical Motor Control (CMC) impairments. This device is based on a proposed architecture that employs advanced technology to evaluate and enhance patients' CMC.

During a pilot study with 10 participants, the device's feasibility and usability were verified, including an initial assessment using the Head Relocation Test and a 12-session intervention over 4 weeks.

The architecture of the proposed system is responsible for gathering pertinent data concerning patients' cervical motor control. It employs advanced algorithms to process this data and objectively assess CMC function. Furthermore, the system tailors the therapy to each patient's individual needs.

Preliminary results indicate that the device and the proposed architecture positively impact assessment test performance accuracy. While additional validation tests are required to confirm their effectiveness, this device emerges as a promising and valuable alternative for assessing and treating patients with CMC impairments. Its focus on advanced technology and personalized adaptation aligns with previous research in telerehabilitation.

4.2 System Architecture

The system uses a multiagent environment that coordinates the flow of information and personalizes therapeutic exercises. For this purpose, it incorporates miniature sensors (Inertial Measurement Units, IMU) that collect real-time movement metrics, correcting deviations through data fusion algorithms.

4.3 Article References

The complete references of the study are: [AbdollahiMKuberPMShiraishiMSoangraRRashediE, AiQLiuZMengWLiuQXieSQ, BarraOrtizHAMatamalaAMInostrozaFLArayaCLMondacaVN, BarzegarKhanghahAFernieGRFekrA, BisioIGaribottoCLavagettoFSciarroneA, BlasHSSMendesASEncinasFGSilvaLAGonzalezGV, CalvaresiDMarinoniMDragoniAFHilfikerRSchumacherM, ChenJOrCKChenT, ChenKBSestoMEPontoKLeonardJMasonAVanderheidenGWilliamsJRadwinRG, DePauwJMercelisRHallemansAMichielsSTruijenSCrasPDHertoghW, DonSDKooningMVoogtLIckmansKDaenenLNijsJ, DuQBaiHZhuZ, GrazioliETranchitaEBorrielloGCerulliCMinganticParisiA, GuidanceM, HidalgoPerezAFGarciaALdUraldeVillanuevaIGMartinezAPAlemanyAFernandezCarneroJLToucheR, HumayunMJhanjhiNZAlmotilagAAlmufarehMF, KazeminasabSNejadghaderiSAAmiriPPourfathiHAKhodaeiMSullmanMJKolahiAASafiriS, KheraPKumarN, KimWSChoSKuJKimYLeeKHwangHJPaikNJ, KumarSRavellaUKShrivastavaAKAnilMSwamySK, LaranjoLDunnAGTongHLKocaballiABChenJBashirRSurianDGallegoBMagrabiFLauAYeal, MalflietAKregelJCoppietersIPauwRDMeeusMRousselNCagnieBDanneelsLNijsJ, MichielsSHertoghWDTruijenSNovemberDWuytsFHeyningPV, MihajlovicZPopovicSBrkicKCosicK, ParkJSJungYJLeeG, PengBYangLLiYLiuTLiuYa, PengBYangLLiYLiuTLiuYb, QuNTianHDMartinoEZhangB, QuNTianHMartinoEDZhangB, RiffittsMOhAAlemuAPatelVSmithCNMuratiSBailesAAllenMDombrowskiMLeeJYeal, RodaCRodriguezACLJaqueroVNavarroEGonzalezP, SpencerJWolfSLKesarTM, SremakaewMJullGTreleavenJUthaikhupS, SulisEMarianiSMontagnaS, TreleavenJa, TreleavenJb, VinoskiS, WeizmanYTiroshOFussFKTanAMRutzE, YoonTLKimHNMinJH, ZoeteRMJOsmotherlyPGRivettDASnodgrassSJ]

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Part IV

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